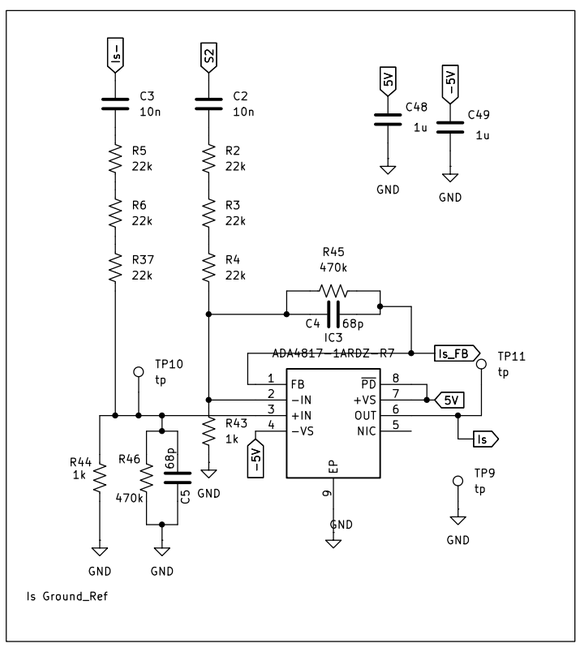
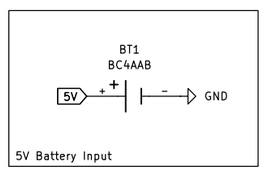
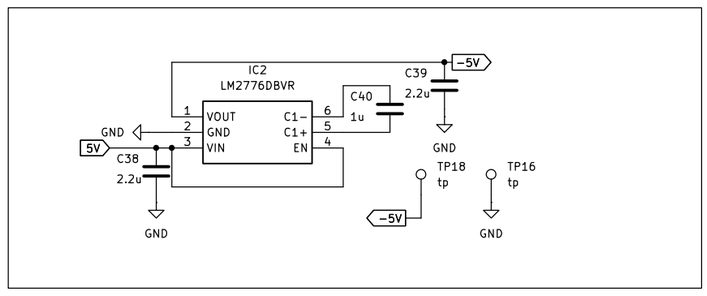
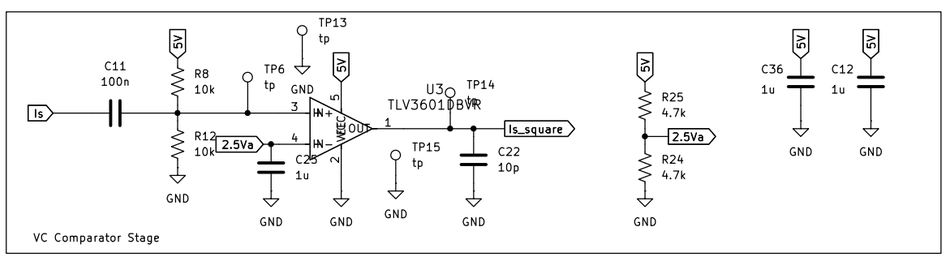
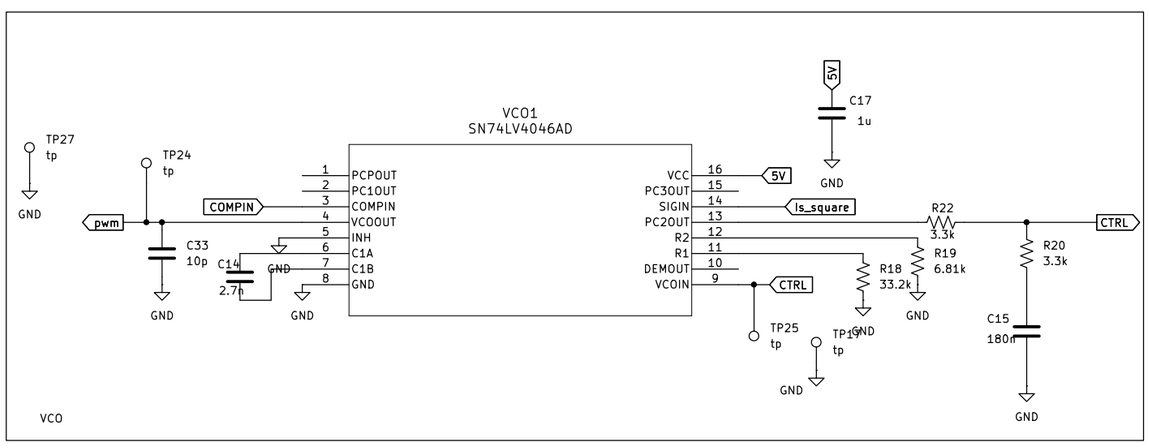

ReceiverCoil1
Screw_Terminal_01x02
33n
33n
C58
C1
C47
1u
Is-
S1
S2
AC Input
TP34
tp
G1
Q1
BSC105N15LS5
TP1
tp
GND
S1
TP19
tp
TP33
tp
TP36
tp
G3
Q2
BSC105N15LS5
TP35
tp
G2
Q3
BSC105N15LS5
TP2
tp
GND
S2
TP22
tp
TP38
tp
TP37
tp
G4
Q4
BSC105N15LS5
TP21
tp
TP5
tp
Rectifier
C27
500u
C19
500u
C54
4.71u
C28
4.71u
C29
4.71u
C30
4.71u
Out+
R32
681k
Vout_s
Active load
R33
47k
C24
47p
GND

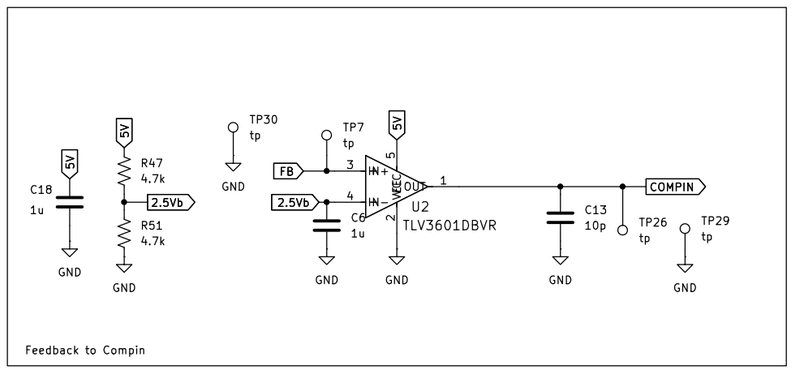

Is-
S2
C3
10n
C2
10n
5V
-5V
C48
1u
C49
1u
GND
R5
22k
R2
22k
R6
22k
R3
22k
R37
22k
R4
22k
R45
470k
C4
68p
IC3
ADA4817-1ARDZ-R7
Is_FB
TP11
tp
TP10
tp
FB
PD
-IN
+VS
+IN
OUT
-VS
NIC
EP
R43
1k
-5V
Is
TP9
tp
R44
1k
R46
470k
68p
C5
Is Ground_Ref

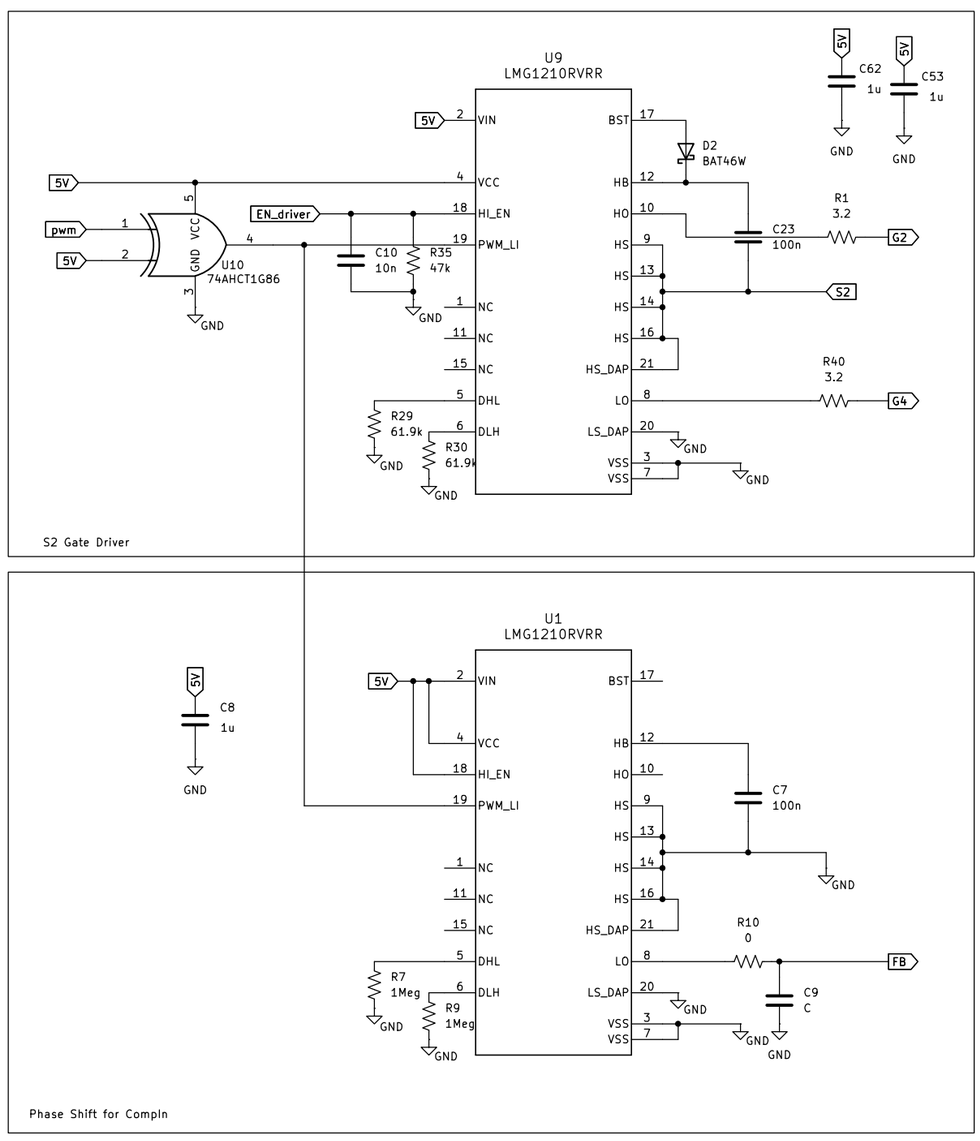

BT1
BC4AAB
5V
GND
5V Battery Input

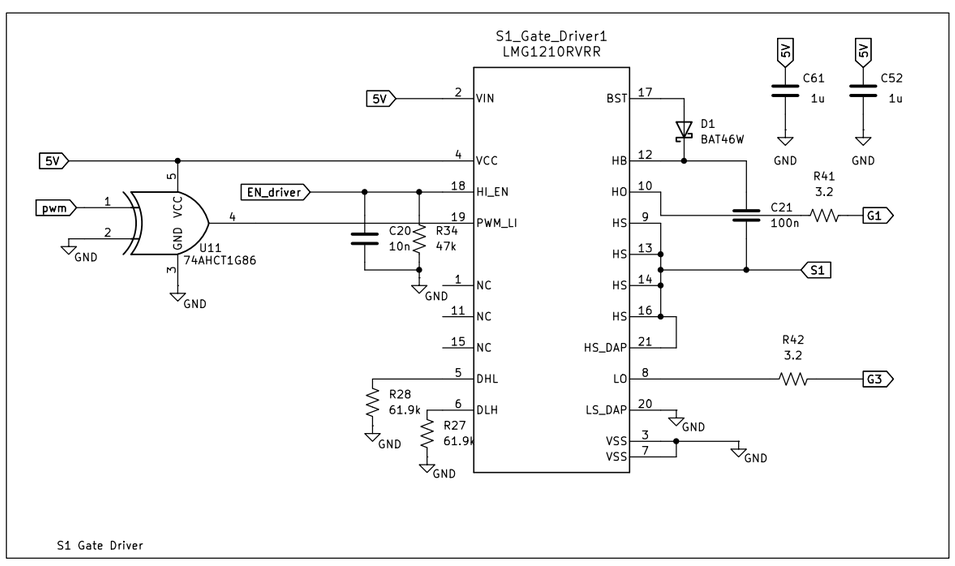

IC2
LM2776DBVR
-5V
C39
2.2u
C40
1u
GND
5V
VOUT
GND
VIN
C1-
C1+
EN
C38
2.2u
TP18
tp
TP16
tp


TP13
tp
C11
100n
5V
R8
10k
TP6
tp
GND
U3
TLV3601DBVR
TP14
tp
Is
R12
10k
2.5Va
C23
1u
TP15
tp
C22
10p
Is_square
R25
4.7k
2.5Va
R24
4.7k
C36
1u
C12
1u
VC Comparator Stage


VCO1
SN74LV4046AD
TP27
tp
TP24
tp
GND
pwm
COMPIN
C33
10p
C14
2.7n
PCPOUT
PC1OUT
COMPIN
VCOOUT
INH
C1A
C1B
GND
VCC
PC3OUT
SIGIN
PC2OUT
R2
R1
DEMOUT
VCOIN
C17
1u
Is_square
R22
3.3k
CTRL
R19
6.81k
R18
33.2k
R20
3.3k
C15
180n
TP25
tp
TP17
tp
VCO


TP30
tp
C18
1u
5V
R47
4.7k
2.5Vb
R51
4.7k
GND
TP7
tp
FB
2.5Vb
C6
1u
IN+
IN-
VCC
OUT
U2
TLV3601DBVR
C13
10p
COMPIN
TP26
tp
TP29
tp
Feedback to Compin


U9
LMG1210RVRR
C62
1u
C53
1u
5V
VIN
BST
D2
BAT46W
VCC
HB
R1
3.2
EN_driver
HI_EN
HO
G2
C23
100n
pwm
U10
74AHCT1G86
C10
10n
R35
47k
PWM_LI
HS
S2
NC
HS_DAP
R40
3.2
DHL
LO
G4
R29
61.9k
DLH
R30
61.9k
LS_DAP
VSS
U1
LMG1210RVRR
C8
1u
C7
100n
R41
3.2
C21
100n
G1
R10
0
FB
R7
1Meg
R9
1Meg
C9
C
Phase Shift for Compln


S1_Gate_Driver1
LMG1210RVRR
C61
1u
C52
1u
5V
VIN
BST
D1
BAT46W
VCC
HB
R41
3.2
EN_driver
HI_EN
HO
C21
100n
G1
pwm
U11
74AHCT1G86
C20
10n
R34
47k
PWM_LI
HS
S1
NC
HS_DAP
R42
3.2
DHL
LO
G3
R28
61.9k
DLH
R27
61.9k
LS_DAP
VSS
S1 Gate Driver